Comparison of Models Forecasting Electricity Dataset

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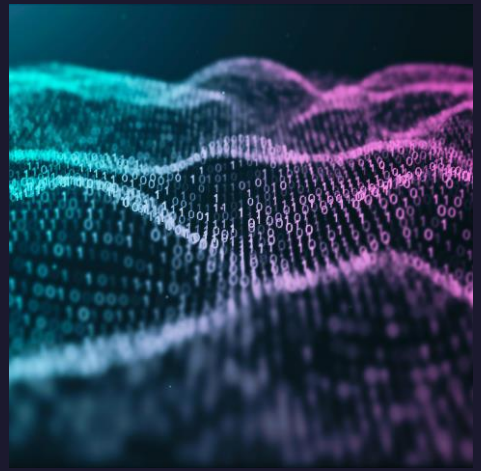


Introduction

We have developed a high-precision forecasting ecosystem that leverages an ensemble of advanced models to deliver 24-hour electricity load predictions for 370 unique grid clients. By moving beyond traditional baselines towards a specialized, cluster-based architecture, this system captures complex consumption patterns with significantly higher accuracy.

Problem

High operational costs and grid instability caused by a "one-size-fits-all" forecasting approach that fails to capture the unique, spiky consumption patterns of diverse consumer clusters.

Objective

Deploy a cluster-optimized forecasting ecosystem to deliver 24-hour predictive intelligence while integrating an Agentic AI interface for real-time stakeholder decision-making.

From:

Static, manual forecasting using legacy baselines like SARIMA/Prophet.

High variance in error (MAPE) during peak holidays and seasonal transitions.

Technical barriers requiring data scientists to interpret results for management.

To:

Specialized, cluster-based modeling using a SOTA Deep Learning ensemble.

Targeted precision that identifies and adapts to unique consumer "Bases" and profiles.

Agentic workflow allowing managers to pull forecasts instantly via natural language.

Saved Operational Costs

Proactive Grid Stability

Strategic Resource Allocation

Process, Standardization and Automation

Data Processing & Clustering

- **Feature Extraction**: Automated calculation of consumption profiles, including weekend/weekday ratios and seasonal strength across 370 clients
- **K-means Clustering:** Implementation of a 4-cluster strategy to group consumers with similar "Bases" and usage patterns
- **Dataset Standard:** Transformation of raw 15-minute readings into hourly aggregated CSVs formatted specifically for Deep Learning ingestion

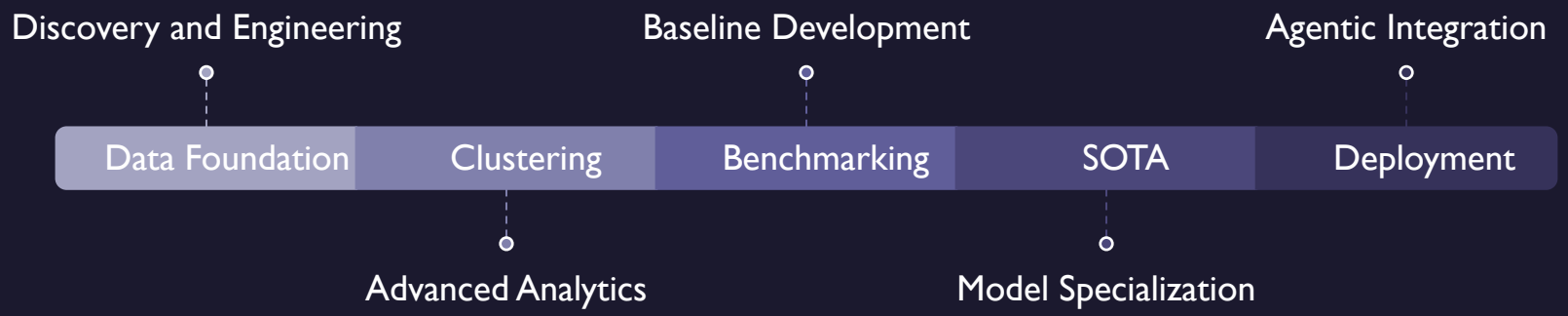
Unified Training Architecture

- **Ensemble Strategy:** Simultaneous training of BasisFormer, PatchTST, and Prophet models to establish a multi-tier performance baseline
- **Evaluation Protocol:** Standardized sliding window evaluation using 0-100 MAPE to ensure cross-cluster comparability
- **Checkpoint Persistence:** Systematically saving model weights to an organized directory for instant deployment

Agentic Automation

- **N8n Orchestration:** Designing a low-code workflow that acts as the switchboard for all incoming manager requests
- **Intelligent Routing:** Scripting the Agent to automatically map Consumer IDs to their corresponding cluster model without human intervention
- **Natural Language UX:** Enables non-technical stakeholders to trigger complex 24-hour forecasting via a simple chat interface

Timeline


Discovery and Engineering
Baseline Development
Agentic Integration
Data Foundation
Clustering
Benchmarking
SOTA
Deployment
Advanced Analytics
Model Specialization

Data

Dataset: UCI Electricity Load Diagrams (2011-2014)

Resampling & Resampling: Converted 15-minute telemetry into 1-hour intervals (26,304 total rows) to focus on macro daily/weekly load cycles and reduce high-frequency noise.

Normalization: We applied StandardScaler to all 370 individual time series aggregated into 3 representative cluster profiles to ensure stable convergence across diverse client scales (residential versus commercial)

Splitting: Excluded 2011 due to data sparsity. Utilized a chronological split (Train: 2012–2013; Test: 2014) to rigorously validate the system's predictive performance on unseen future data.

Clustering

We grouped the 370 clients into 4 behaviorally distinct clusters using their hourly consumption shapes, weekday/weekend patterns, and seasonality strength.

- **Cluster 0 — Residential** (215 clients): clear morning & evening peaks
- **Cluster 1 — Industrial** (35 clients): sustained high loads, shift-schedule driven
- **Cluster 2 — Mixed Use** (91 clients): high variance, irregular patterns
- **Cluster 3 — Smooth Baseload** (29 clients): flat, minimal daily fluctuation

Each cluster gets a model tuned to its behavior. This directly translates to:

- Lower forecast error on high-value industrial accounts (Cluster 1 GB: 8.63% MAPE)
- Accurate 24hr forecasts translate to less over-purchasing of reserves
- New clients are automatically assigned to the right model at onboarding

Baseline Models

Baseline Models

Models: We wanted to experiment with some techniques we covered in class

- Seasonal Naive
  - Predicts future value from last observed value in the same season
- SARIMA
  - Captures short term autocorrelation & repeating daily patterns
- Prophet
  - Decomposes time series into:
    - trend g(t)
    - seasonality s(t)
    - holiday effects h(t)
  - Acts as interpretable baseline with flexible trend modeling

Results

Both Prophet and SARIMA underperform relative to the seasonal naive baseline across most clusters, highlighting the limitations of traditional statistical approaches on this dataset. All baseline models struggle particularly in high-variance clusters, motivating the use of more flexible approaches.

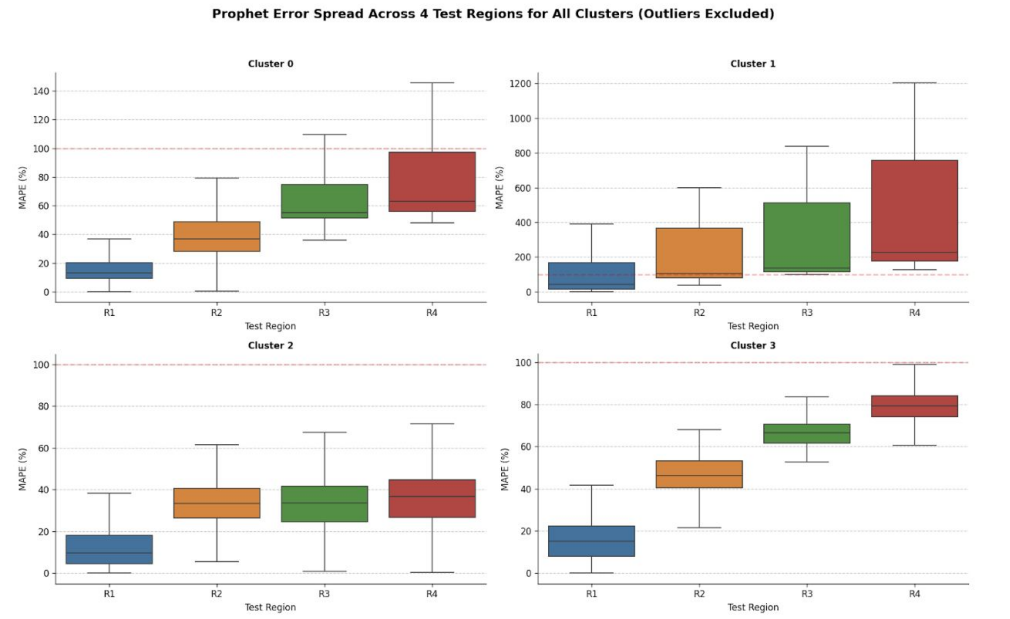


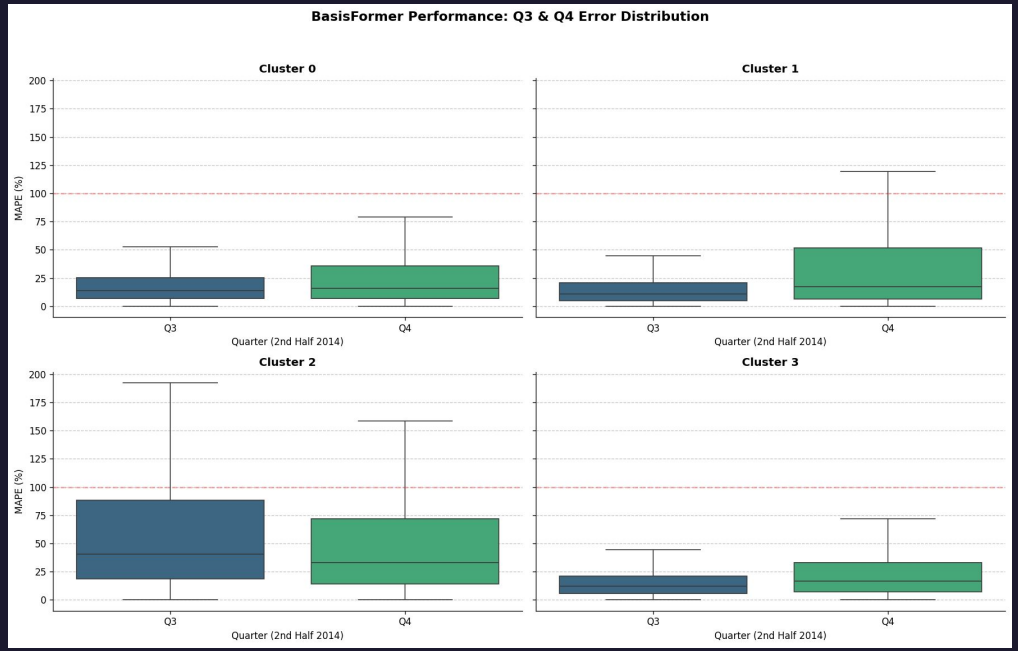
Figure 12: MAPE values for Prophet presented per cluster profile

Improved Models

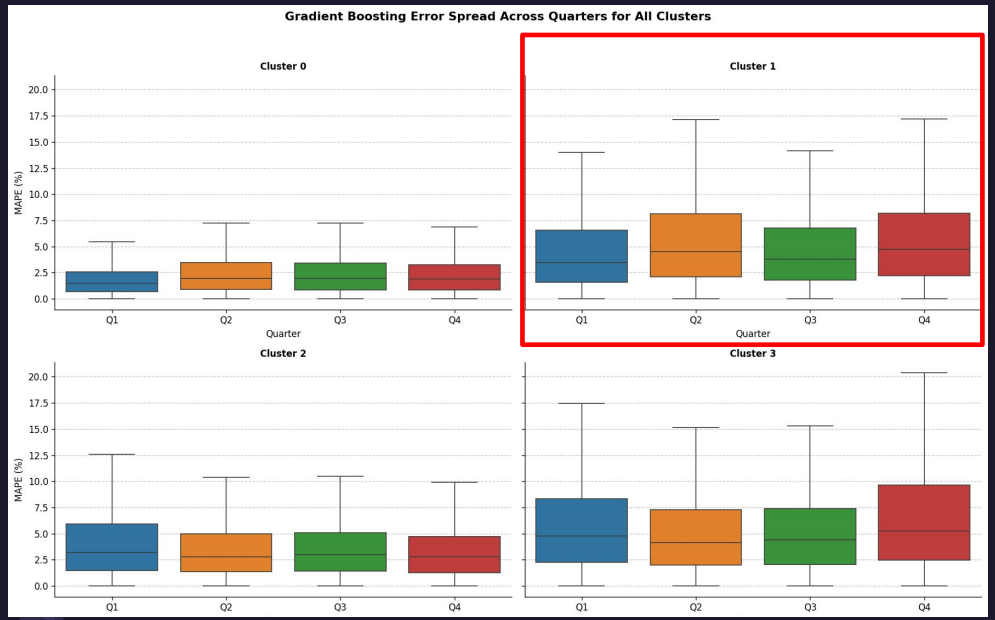
BasisFormer Results

Basis Former shows decent results for residential and industrial clusters

BasisFormer represents a paradigm shift in our forecasting ensemble, moving away from the rigid seasonality of traditional statistical models toward a shared, learnable basis space able to manage the irregular industrial grid.

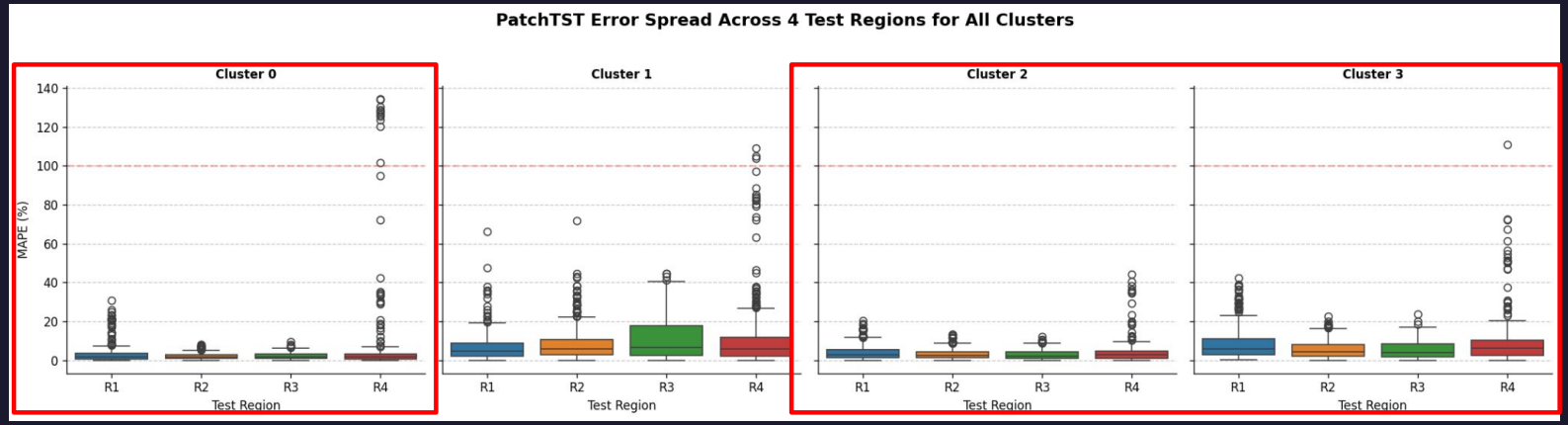


GB Results



- **Gradient Boosting wins on Cluster 1 (Industrial)**
  - Cluster 1 holds the highest-consumption and highest-cost clients
  - 8.63% MAPE on industrial accounts directly reduces over-procurement of energy reserves; ultimately reducing costs

PatchTST Results



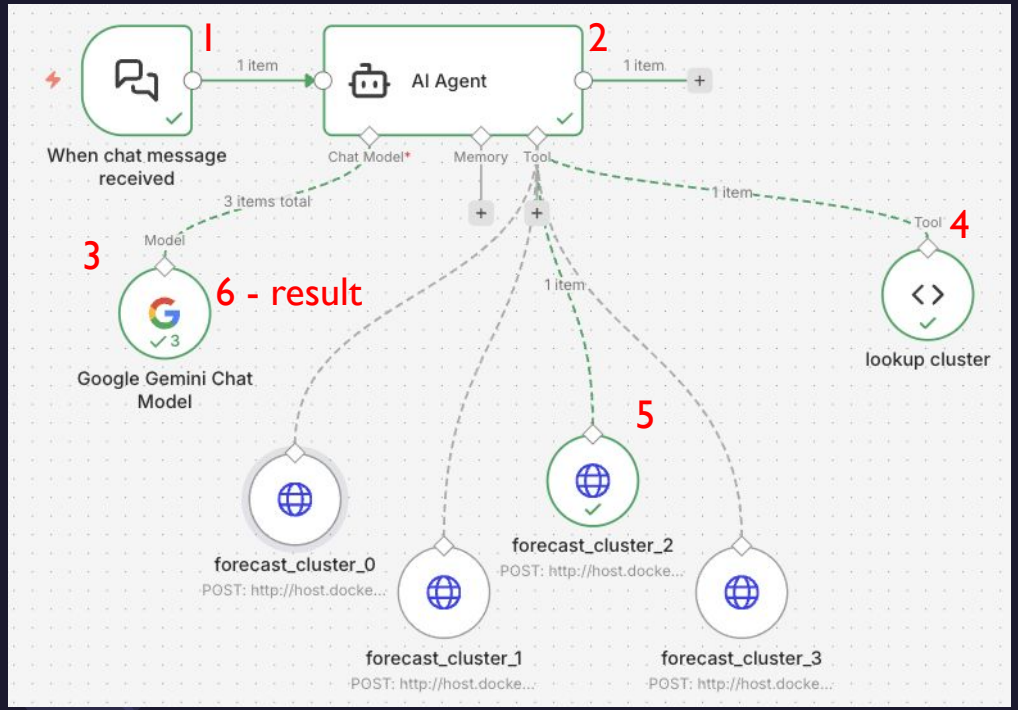
- **PatchTST leads on Residential, Mixed, Smooth Baseload clusters**
  - PatchTST's ability to learn complex seasonal patterns without manual feature engineering makes it the strongest general-purpose model

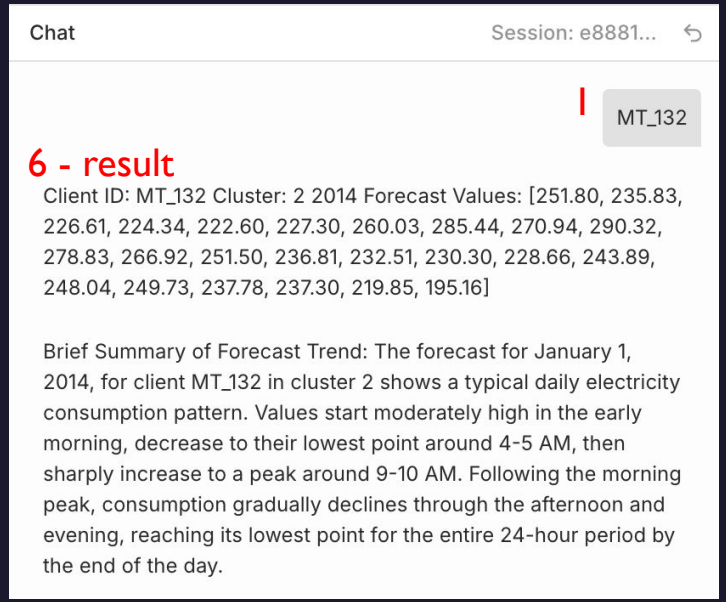
Model Comparisons - MAPE scores

	Seasonal Naive	SARIMA	Prophet	Basis-Former*	Gradient Boost (GB)	PatchTST
Cluster 0 (Residential)	6.41%	29.15%	50.26%	171.99%	4.59%	4.49%
Cluster 1 (Industrial)	11.16%	45.34%	258.94%	107.88%	8.63%	12.18%
Cluster 2 (Mix-use)	15.11%	136.6%	42.64%	136.44%	14.59%	13.60%
Cluster 3 (Critical Infrastructure)	117.41%	1227.37%	121.18%	432.39%	110.73%	110.17%

Please see technical report for more detailed information on specific models featured here

n8n Agent Implementation





Challenges & Limitations

Data & Metric Sensitivity

- **The Challenge:** High percentage errors (up to 1500%) were observed in Region 4 (Q4 2014).
- **The Insight:** This was largely a mathematical artifact of the MAPE metric. When actual consumption drops near zero (standby power), even a physically tiny prediction error (e.g., 0.5 kWh) results in a massive percentage spike.
- **The Limitation:** Our current models are "univariate," meaning they only see past load data and lack the context to know why a consumer suddenly went on vacation or reduced usage.

Model Generalization

- **The Challenge:** All models (BasisFormer, Prophet, PatchTST) showed increased variance during the November–December holiday season.
- **The Insight:** Electricity consumption in the Portuguese grid is highly sensitive to exogenous variables like sudden temperature drops or public holiday shifts which are not captured in the raw timestamp.
- **The Limitation:** Without Weather Data (Temperature/Humidity) or a Holiday Calendar, the models treat these spikes as "noise" rather than predictable patterns.

Future Roadmap

- **Feature Expansion:** Integrate Weather API data and national holiday flags to stabilize Q4 predictions. Incorporate more context for better predictions and more cohesive story.
- **Hybrid Loss Functions:** Move toward a custom loss function that penalizes large absolute errors more than percentage errors during low-load windows.
- **Agentic Feedback Loop:** Implement a "Human-in-the-Loop" trigger in n8n that flags forecasts with high uncertainty for manual review by the grid manager.

Summary

This project implements a sophisticated ensemble of forecasting models to deliver precise 24-hour load predictions tailored to diverse consumer behavioral clusters. By significantly reducing predictive error in high-volatility sectors, the framework generates tangible business value through optimized energy procurement and minimized operational costs for spinning reserves. These technical insights are finalized within a front-facing, agentic AI application that transforms complex telemetry into an interactive, natural-language decision-support tool for real-time grid management.

Team

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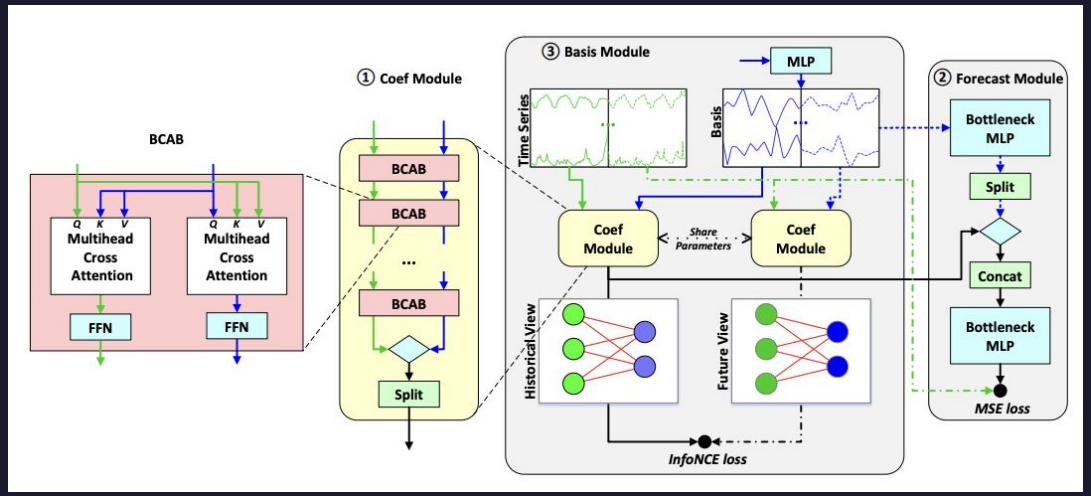
Architecture - BCAB Block



Basis Module

- A self-supervised module that learns the library of fundamental patterns (Bases) directly from the 370-client historical data

Coefficient Estimation

- Instead of predicting values, the model calculates "coefficients" to determine how to mix the learned bases to match the 96-hour input window

Forecast Module:

- Utilizes the learned bases and coefficients learned in the previous step to create an accurate forecast prediction perfect for the periodicity of the Electricity data